

Housing, Fertility, and Demographic Equilibrium

A two-generation model

Environment

Agents. Time is discrete and one period is one generation. At date t , the economy contains Y_t young households and O_t old households. A young household has income y_i and liquid wealth b_i , drawn from a fixed type distribution F . A young renter becomes an old renter, a young owner becomes an old incumbent, and every old household exits at the end of the period.

Preferences. A young household values adult consumption c , space net of children's needs, and completed fertility:

$$u_t^Y(c, h, n) = \log c + \alpha \log(h - \kappa n) + \vartheta_t \log n. \quad (1)$$

Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of space. An old household values consumption, retained housing, and the estate delivered at death:

$$u_t^O(c^O, h^O, e) = \log c^O + \gamma \log h^O + \omega_B \log e. \quad (2)$$

The financial sector supplies liquid claims and absorbs estates at price q .

Housing and financial markets. A fixed stock $\bar{H}$ produces one unit of housing services per unit and period. Renters can occupy at most $\bar{h}^R$, while owners can occupy at most $\bar{h}^O > \bar{h}^R$. The asset price is P_t , one-period-ahead goods have price $q \in (0, 1)$, and $R = q^{-1}$. Competitive housing intermediaries imply the user-cost relation:

$$r_t = (1 + \tau_t^p)P_t - qP_{t+1}, \quad (3)$$

where $\tau_t^p P_t$ is the recurring property tax per unit.

Ownership and taxation. A buyer finances the share ϕ of the purchase with a one-period mortgage and posts $(1 - \phi)P_t h$ at closing. The mortgage advance is $\phi P_t h$ and its next-period repayment is $R\phi P_t h$. Selling a unit realizes the capital-gains tax

$$\ell_t = \tau^g \left[P_t - \frac{P_{t-1}}{1 + \varpi} \right]_+, \quad (4)$$

while a unit retained until death enters the estate with stepped-up basis. These cash flows give two tenure-specific costs. Buying a unit, paying its property tax, and selling it next period costs:

$$r_t^O = (1 + \tau_t^p)P_t - q(P_{t+1} - \ell_{t+1}) = r_t + q\ell_{t+1}. \quad (5)$$

For an old incumbent, retaining a unit rather than selling it, paying the property tax, and placing its next-period value in the estate costs:

$$r_t^I = (P_t - \ell_t) + \tau_t^p P_t - qP_{t+1} = r_t - \ell_t. \quad (6)$$

Thus the buyer anticipates the future tax, while the incumbent avoids the current tax by retaining the house.

Demography. Each child generates $\nu > 0$ future young households after survival and household formation. The children of a household affect future cohort size but do not inherit that household's individual state.

Household problems

Old households

An old incumbent enters with net financial wealth a and a house H . In current-value terms, selling the entire house produces $(P_t - \ell_t)H$; retained housing costs $r_t - \ell_t$ by equation (6). Her problem is therefore:

$$\begin{aligned} V_t^O(a, H) &= \max_{c^O, h^O, e} u^O(c^O, h^O, e) \\ \text{subject to } &c^O + qe + (r_t - \ell_t)h^O = a + (P_t - \ell_t)H + T_t, \quad 0 < h^O \leq H, \quad e \geq P_{t+1}h^O. \end{aligned} \quad (7)$$

The estate includes the next-period value of retained housing. When its composition constraint and $h^O \leq H$ are slack, the incumbent's marginal value of retained space is:

$$\frac{u_h^O}{u_c^O} = \frac{\gamma c_t^O}{h_t^O} = r_t - \ell_t. \quad (8)$$

An old renter has no house or gain and solves the same problem with budget $c^O + qe + r_t h^O = a + T_t$.

Young households

Conditional on renting, a young household chooses current consumption, housing, fertility, and saving:

$$W_t^R(i) = \max_{c,h,n,a'} u_t^Y(c,h,n) + \beta V_{t+1}^R(Ra') \quad (9)$$

subject to $c + \chi n + a' + r_t h = y_i + b_i + T_t, \quad 0 < h \leq \bar{h}^R, \quad a' \geq 0.$

Conditional on owning, the same household purchases h units, posts the down payment at the beginning of the period, and repays the mortgage when old:

$$W_t^O(i) = \max_{c,h,n,a'} u_t^Y(c,h,n) + \beta V_{t+1}^O(a_{t+1}, h) \quad (10)$$

subject to $c + \chi n + a' + [(1 - \phi) + \tau_t^p] P_t h = y_i + b_i + T_t,$
 $a_{t+1} = Ra' - R\phi P_t h, \quad (1 - \phi) P_t h \leq b_i,$
 $0 < h \leq \bar{h}^O, \quad a' \geq 0.$

Income received after closing can finance consumption and saving but cannot be pledged for the down payment. The household chooses the tenure with the higher value.

Suppose saving is interior and the future old retention limit is slack. Let λ_t^m be the multiplier on the young budget, μ_t the multiplier on the owner's down-payment constraint, and η_t^m the multiplier on the tenure-specific size limit. The access values in current goods are:

$$\zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \quad \zeta_t^O = \underbrace{\frac{\mu_t}{\lambda_t^O} (1 - \phi) P_t}_{\zeta_t^{O,F}} + \underbrace{\frac{\eta_t^O}{\lambda_t^O}}_{\zeta_t^{O,S}}. \quad (11)$$

Using the saving Euler equation and the old value-function envelopes, the housing conditions reduce to:

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_t^m, \quad r_t^R = r_t, \quad r_t^O = r_t + q\ell_{t+1}. \quad (12)$$

The ownership expression follows from the purchase, mortgage, property-tax, and resale cash flows in equation (10). The fertility condition prices the goods and space required by a marginal child:

$$\frac{\vartheta_t c_t^m}{n_t^m} = \chi + \kappa(r_t^m + \zeta_t^m). \quad (13)$$

A housing constraint raises the household's shadow price of a child by $\kappa\zeta_t^m$.

Equilibrium and steady states

Let $\bar{n}_t$, $\bar{h}_t^Y$, and $\bar{h}_t^O$ denote fertility and housing averaged over the young type distribution and the inherited old distribution. Let $\bar{s}_t^O$ denote housing sold by an average old incumbent. Cohort sizes and the housing stock obey:

$$Y_{t+1} = \nu \bar{n}_t Y_t, \quad O_{t+1} = Y_t, \quad Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}. \quad (14)$$

The government rebates property- and capital-gains-tax receipts equally across living households:

$$(Y_t + O_t) T_t = \tau_t^p P_t \bar{H} + \ell_t O_t \bar{s}_t^O. \quad (15)$$

This account makes both taxes transfers while preserving their effects on private timing and housing choices.

Definition 1 (Transition equilibrium). Given the initial cohort masses and old distribution, a transition equilibrium is a sequence of prices, transfers, household policies, distributions, and cohort masses such that households solve equations (7)–(10); the old distribution is generated by the preceding young choices; housing clears and the government budget balances at every date; equation (3) prices the housing asset; equation (14) advances the cohorts; and the asset-price path satisfies the no-bubble condition $\lim_{J \rightarrow \infty} q^J P_{t+J} = 0$.

A positive closed steady state is a constant transition equilibrium with $Y^* = O^* > 0$. Let $\mathcal{N}(P, T)$, $\mathcal{H}^Y(P, T)$, $\mathcal{H}^O(P, T)$, and $\mathcal{S}^O(P, T)$ denote the stationary aggregate policies generated by the household problems. The stationary values satisfy:

$$1 = \nu \mathcal{N}(P^*, T^*), \quad (16)$$

$$\bar{H} = Y^* [\mathcal{H}^Y(P^*, T^*) + \mathcal{H}^O(P^*, T^*)], \quad (17)$$

$$2Y^* T^* = \tau^p P^* \bar{H} + \ell(P^*) Y^* \mathcal{S}^O(P^*, T^*). \quad (18)$$

At a constant real price, $\ell(P^*) = \tau^g P^* \varpi / (1 + \varpi)$. The three conditions jointly determine the stationary price, transfer, and population scale. Replacement supplies the restriction absent from a stationary cross-section with an exogenously fixed cohort mass.

Proposition 1 (Existence of a positive steady state). *Suppose equations (17)–(18) define continuous functions $\hat{Y}(P) > 0$ and $\hat{T}(P)$ on the feasible price set. Define the renewal schedule $\Gamma(P) = \nu \mathcal{N}(P, \hat{T}(P))$. A positive closed steady state exists if and only if $\Gamma(P) = 1$ at a feasible price. It is unique if Γ crosses one exactly once.*

Proof. Stationarity of the aging law gives $O^* = Y^*$. The young-cohort law then gives $Y^* = \nu \mathcal{N}(P^*, T^*) Y^*$, which is equivalent to equation (16) for $Y^* > 0$. The remaining two equations determine the scale and rebate. The converse follows by constructing the constant sequence from any feasible root. $\square$

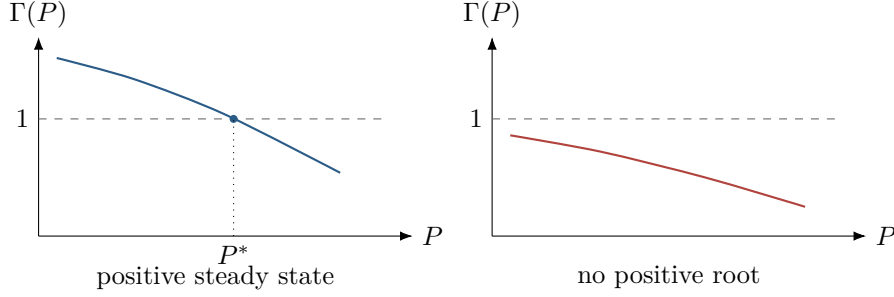


Figure 1: The renewal schedule is induced by household fertility, housing clearing, and the balanced rebate. The monotone schedules shown illustrate a unique positive steady state and a case with no positive root.

After a permanent change in fertility preferences, the inherited cohorts and old distribution are fixed on impact. Prices and household choices adjust, fertility changes the next young cohort, and the new cohort changes subsequent housing demand. A path converges to another positive steady state only when a replacement root exists and the equilibrium dynamics select it. When $\Gamma(P) < 1$ throughout the feasible price set, the closed economy instead follows a nonstationary contraction path.

Intergenerational housing allocation

Consider an old incumbent at an interior retention choice and a young owner whose down-payment constraint binds while the owner-size limit is slack. Their marginal values of current housing services are:

$$MV_{it}^Y = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}, \quad MV_{jt}^O = r_t - \ell_t. \quad (19)$$

The market component cancels when a unit is transferred between them.

Proposition 2 (Compensated intergenerational reallocation). *Hold the price sequence, fertility, cohort masses, tenure assignments, and saving choices fixed. Suppose an institution can transfer one marginal unit of existing housing from household j to household i , compensate all affected households and the government, and incurs real cost K_{ijt} per unit. The goods-equivalent surplus is:*

$$\mathcal{S}_{ijt} = \zeta_{it}^{O,F} + q\ell_{t+1} + \ell_t - K_{ijt}. \quad (20)$$

The equilibrium allocation fails the conditional housing-allocation test whenever $\mathcal{S}_{ijt} > 0$.

Proof. Increase the young household's housing by dh and reduce the incumbent's retained housing by the same amount. Compensating the two households at their marginal utilities leaves $(MV_{it}^Y - MV_{jt}^O - K_{ijt})dh$ units of goods. Substituting equation (19) gives equation (20). Equation (15) returns the two tax payments to the household sector, so they are private wedges rather than resource costs. $\square$

The reallocation result and the fertility response are distinct. Let $A = 1 + \gamma + \omega_B$ and $k = \beta A$. With interior saving and old choices, substituting the old allocation and the saving Euler equation into either young problem gives the lifetime resource identity:

$$(1 + k)c + \chi n + r_t^m h = \mathcal{W}_t, \quad \mathcal{W}_t = y_i + b_i + T_t + qT_{t+1}. \quad (21)$$

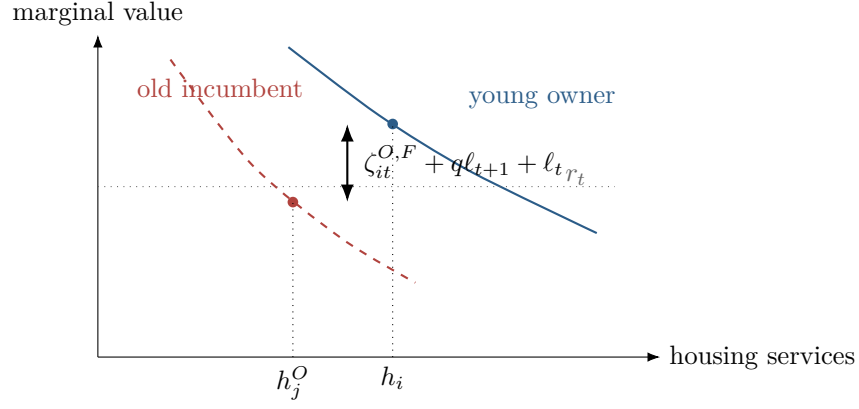


Figure 2: A constrained young owner values a marginal unit above the market user cost, while the incumbent values retained space below it. The vertical distance is the reallocation surplus before real implementation cost.

Suppose an access constraint fixes the household's housing at $\hat{h}$ while fertility remains interior. Holding $\mathcal{W}_t$ and prices fixed, implicitly differentiating the fertility condition yields:

$$\frac{dn}{d\hat{h}} = \frac{\frac{\alpha\kappa}{(\hat{h} - \kappa n)^2} - \frac{(1+k)\chi r_t^m}{(\mathcal{W}_t - r_t^m \hat{h} - \chi n)^2}}{\frac{\vartheta_t}{n^2} + \frac{(1+k)\chi^2}{(\mathcal{W}_t - r_t^m \hat{h} - \chi n)^2} + \frac{\alpha\kappa^2}{(\hat{h} - \kappa n)^2}}. \quad (22)$$

The first term in the numerator is the value of additional family space; the second is the consumption cost of occupying more housing. Equation (12) implies $[\mathcal{W}_t - r_t^m \hat{h} - \chi n]/[\hat{h} - \kappa n] = (1+k)(r_t^m + \zeta_t^m)/\alpha$. Since $\zeta_t^m \geq 0$, a sufficient condition for greater housing access to raise fertility is:

$$\chi \leq \frac{(1+k)\kappa r_t^m}{\alpha}. \quad (23)$$

Under this condition, an institution that reaches a positive financing gap and increases the household's housing both improves the conditional allocation and raises fertility among the constrained households it directly affects. General-equilibrium prices, rebates, and cohort sizes then propagate that response through equations (14)–(15).